Answers and talk notes

Divide Mixed Numbers

What your student is learning

Tonight we divide a mixed number by a fraction to answer a real cutting problem — how many pieces a length gives.

Answers

Ask your student to explain before you look.

1. **B.** Change 3 1/4 to an improper fraction. Change 3 1/4 to an improper fraction first: 3 1/4 = 13/4.
2. **D.** 5 1/4 ÷ 3/4. The whole rope is split into 3/4-foot pieces, so 5 1/4 ÷ 3/4.
3. **B.** 4. 2 1/2 = 5/2, and 5/2 × 8/5 = 40/10 = 4.
4. **A.** 7. 5 1/4 = 21/4, and 21/4 × 4/3 = 84/12 = 7.
5. **C.** 3. 3 3/4 = 15/4 and 1 1/4 = 5/4, so 15/4 × 4/5 = 60/20 = 3 pieces.
6. **C.** about 17. Each whole holds 2 halves, so 8 1/2 wholes hold about 17 halves.

Talk about it

Did you change the mixed number into an improper fraction before dividing? How many pieces make sense?

Home mission

Cut a paper strip that is 6 inches long. Ask: how many pieces that are each 1 1/2 inches long can you cut? Guess first, then fold or cut to check.

When they are stuck, ask

- Is the mixed number a single fraction yet?
- How do we turn a mixed number into an improper fraction?
- How many of those pieces fit into the whole length?

Try not to say

- Divide the whole numbers, then the fractions.
- Leave it as a mixed number and just flip it.
- A leftover piece does not need to be mentioned.